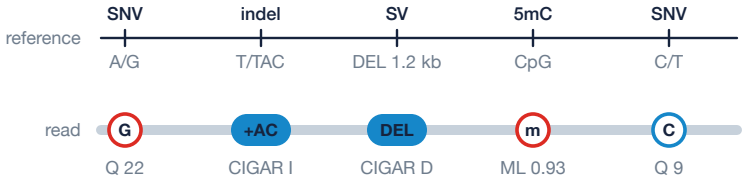


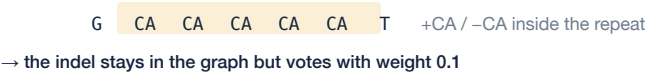
a Allele observations from one alignment



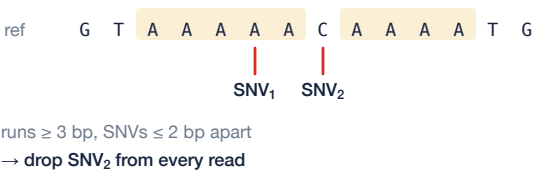
observation quality entering the edge weight

class	source	weight
SNV	aligned base, Phred Q	1 if both bases $\geq$ Q12, else 0.1
indel	CIGAR I/D at the site	1; 0.1 if in a tandem repeat
SV	caller read list	1 for ALT, 0.1 for REF
5mC	modcall read lists	1, strand matched

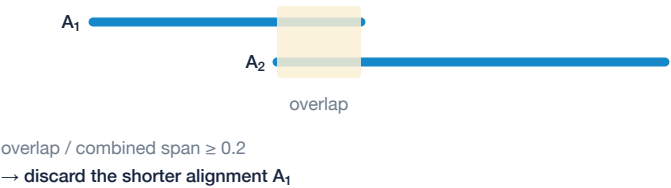
d Tandem-repeat indel flag



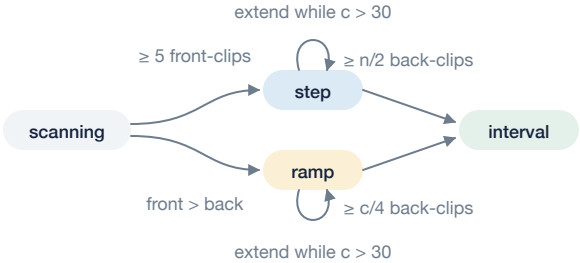
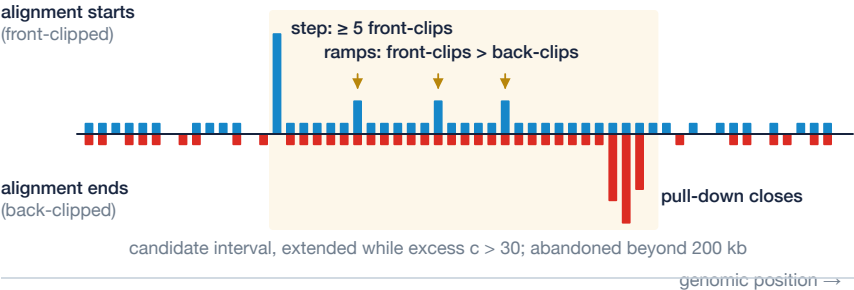
b Homopolymer SNV filter (nanopore, --ont)



c Overlapping-alignment filter



e Copy-number-aware filter: two-state detection of copy-number-altered intervals

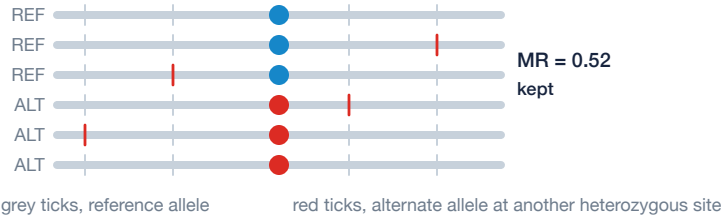


f Mismatch load separates true heterozygotes from paralogues

$$MR = \frac{\text{mean } R_m(\text{ALT})}{\text{mean } R_m(\text{REF}) + \text{mean } R_m(\text{ALT})} \geq 0.7 \rightarrow \text{remove the site from every read}$$

R_m = alternate alleles a read carries at other sites inside the interval

true heterozygous site



parologue-specific difference



order of operations

